Internal Combustion Engine Lab

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ME 4056 - Section L01

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Introduction

Burdell, Inc. has been contracted by Generative Energy Systems, Inc. to determine the mechanical parameters of a new modular combustion engine and develop a control system to control the response. This modular system was developed with the capacity to process air and propane fuel sources, as well as the coupling of a 48V regenerative motor capable of driving the motor or receiving power from the motor. After calibration of the system, the outcome of this report is to determine the engine's mechanical properties, and model engine output with simulated combustion.

Methods

The test motor system is described in Figure 1. Mass was added incrementally to the aperture arm to calibrate the force transducer voltage output. Calibration tests were reinitialized following subsequent days in the laboratory. Once the constants of the calibration equation were determined, physical quantities of the engine such as piston dimensionality, top-dead-center location, and encoder accuracy were quantified and recorded. Initial experimentation of the engine with four flywheels included the measurement of motor torque and piston location against the encoder angular output across various angular velocities. The output data was numerically integrated to obtain the work done by the motor on the engine during the steady state and transitional phases. This process was then repeated with only one flywheel attached to the engine across a single, stable angular velocity. A complete mathematical model was created using the data collected from the experimentation to represent the work-kinetic energy relationship of the engine.

Experimentation of the engine continued with the inclusion of an engine head and four flywheels without air injection. Once in steady-state, angular data was measured at each speed for approximately three seconds. With the head on, torque values from the force transducer were also collected for analysis against time. Correlating the two sets of data, numerical integration of the torque at each angular position provided the work done by the motor on the engine. This process was repeated without the inclusion of any flywheels. Documentation of work done by and on the engine was updated to include the results of this experimentation phase. Finally, uncertainty values of the torque, angular position, and work done by the motor were calculated and documented.

The final phase of experimentation models the effects of air injection to simulate the ignition of fuel in the power stroke. Theoretical power stroke angular position values were digitally input into the *LabVIEW* interface to coordinate the boundary conditions of air injection. This digital signal was then converted into an analog signal via the *NI ELVIS III* control board. This analog signal was fed through a flip-flop switch to differentiate the angular superposition of the power stroke and the intake stroke. Tuning of the boundary conditions digitally was undergone to optimize work output by the engine on the motor. Once completed, the work-energy model was updated to include the results of the air injection.

Results and Discussion

To calculate the experimental offset of the motor setup, known applied torques and their associated voltage output from a force transducer can be represented by the calibration equation

of $T(V) = 0.64882V + 0.75802$ for the first and second set of experiments, illustrated by Figure 2. Recalibration data, generated subsequently for air injection testing, is represented by the calibration equation $T(V) = 0.84769V + 1.2925$, also featured in Figure 2. Following this, the motor was run at low, medium, and high speeds with all flywheels attached; the motor's angular position encoder data and cumulative encoder angle are plotted against time from start until the motor's speed reached steady state as seen in Figure 3. Using the torque calibration equation and the collected force transducer data, the motor torque is plotted against the incremental angular position in Figures 4, 5, and 6 for low, medium, and high speeds. The resulting data is very noisy, so a rolling average of the torque data is calculated and plotted alongside the collected torque data. The same data is shown at high speed with only 1 flywheel, plotted against the data with 4 flywheels in Figure 7, showing that significantly less motor work is produced with only 1 flywheel.

The work done by the motor on the engine is calculated using part of the governing work-energy balance equation (Equation 1) by taking the integral of the averaged torque over time (Equation 2) as plotted in Figures 8, 9, and 10 for the low, medium, and high speeds respectively. The work of friction is the work done by the motor under its steady state rolling average torque (Equation 2). The energy of the flywheels, calculated using the governing work-energy balance equation (Equation 1), is simplified to only have friction and motor work components for the initial experimentation without a motor head to produce air work. The flywheel energy is the difference between the work produced by the motor and friction work, shown in the transient region where the motor has not yet reached steady state. The flywheel energy can be used to calculate the experimental inertias of the flywheels and power produced by the motor using Equation 2, shown in Table 1. The experimental flywheel inertias were significantly lower than the theoretical values, decreasing as the motor speed increased rather than staying constant, despite inertia being a time-invariant, physical property. This is due to discrepancies in the work caused by additional inertial elements such as the bearing or the motor shaft. The power produced by the motor increased as the speeds increased, as expected, going from 66.5 W at low speed with 4 flywheels to 885.5 W at high speed with 4 flywheels, and decreasing to 181.95 W at high speed with only 1 flywheel. This is expected, as greater speeds and higher inertias require more power to be moved.

With the engine head on, the rolling average torques are calculated for low-speed data sets with all and no flywheels attached due to higher reliability at lower speeds. The resulting motor work is calculated for each stroke of the 4-stroke cycle, illustrated in Figures 11 and 12. The experimental air work is calculated using Equation 3 and compared to the theoretical air work value for each stroke is calculated using Equation 4. The experimental uncertainty of the work of each stroke was calculated using the equations shown in Equation 5. These values are compared for each stroke at a low RPM with no flywheels attached in Figure 13 as shown in Table 2. The intake and exhaust strokes are not theoretically supposed to generate any work, and experimentally the intake stroke generated a significant amount of work of 1.233 J, with the theoretical work falling outside the uncertainty bounds of ±0.0886 J, while the exhaust stroke generated some work, -0.065 J, however an insignificant amount that catches the theoretical work value within its uncertainty bounds of ±0.0885 J. The compression and power strokes produced significantly less work than the motor theoretically should have, producing -5.46 J and 5.23 J respectively, compared to a theoretical produced work of -18.959 J and 18.959 J, which fall far out of the experimental

uncertainty bounds. This large discrepancy is likely due to imperfections in the model IC engine that decrease the work produced by the air in the engine head as well as faulty force transducers that collected noisy data as seen in Figures 4, 5, and 6.

A linear fit is created to determine the angular velocity of the engine at steady state for each different set of start and end injection angles, as shown in Figure 14 for a 95° start and 145° end injection angle. Air injection periods of less than 180° were selected since air injection periods greater than 180° would lead to the air doing negative work and slowing down the motor. The work for each stroke of a 4-stroke cycle at steady state is modeled using Equation 4, shown in Figure 15 for a 100° start and 155° end injection angle. The full results for the motor work created by each start and end injection angle parameter set can be seen in Figure 16 with data shown in Table 3, alongside the experimental uncertainty for the motor work, calculated using Equation 5. The 105°-165° start and end injection angles created the highest motor work, 29.217 J. The theoretical work that the system should produce for each stroke and set of injection angles is shown in Table 4, showing the work discrepancy between the experimental and model values, calculated using Equation 6. There are large discrepancies in the data unaccounted for in the calculated uncertainty margins, specifically in the power and compression strokes. These large discrepancies are likely due to issues with the motor model itself or the sensors used during the experiment, similar issues being seen in the un-averaged data in Figures 4, 5, and 6, which may have amplified in the gap between experiments creating unusable data.

Conclusion

For Generative Energy Systems to maximize the work of the motor of their modular internal combustion engine, Burdell, Inc. recommends beginning air injection at 105 degrees and to stop air injection at 165 degrees based on compressed air analysis. It is important to not interfere with the other strokes and to absolutely avoid ever choosing an angle injection period that exceeds 180 degrees. Given the presence of a large $W_{discrepancy}$, the theoretical model was oversimplified and therefore it is not accurate to a physically realized system. For instance: there is no net work done in the exhaust and intake strokes of the theoretical model. This can be attributed to issues in the experimental motor, such as issues with the sensors, specifically the force transducer, and opposing forces to motor due to friction and accidental air injection during the exhaust phase. Future work investigating the improvement of force transducer sensor input and system stability during fuel injection may reduce such discrepancies between the theoretical and the experimental. Additional sensor readings of intake pressure for equalization of fuel injection flow may also help produce a more accurate model of the internal combustion engine.

Appendix:

Figures:

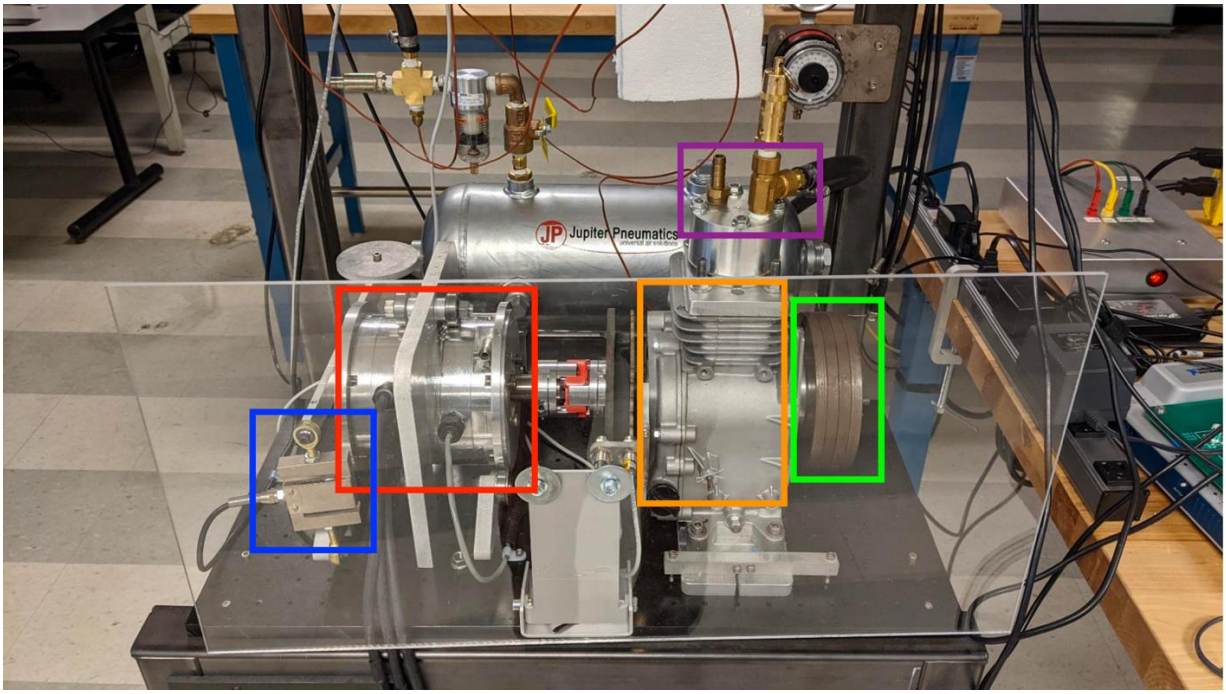


Figure 1: Experimental Setup With the Engine (Orange Box), Motor (Red Box), Flywheels (Green Box), Air Injection Input (Purple Box), and Force Transducer (Blue Box) [1]

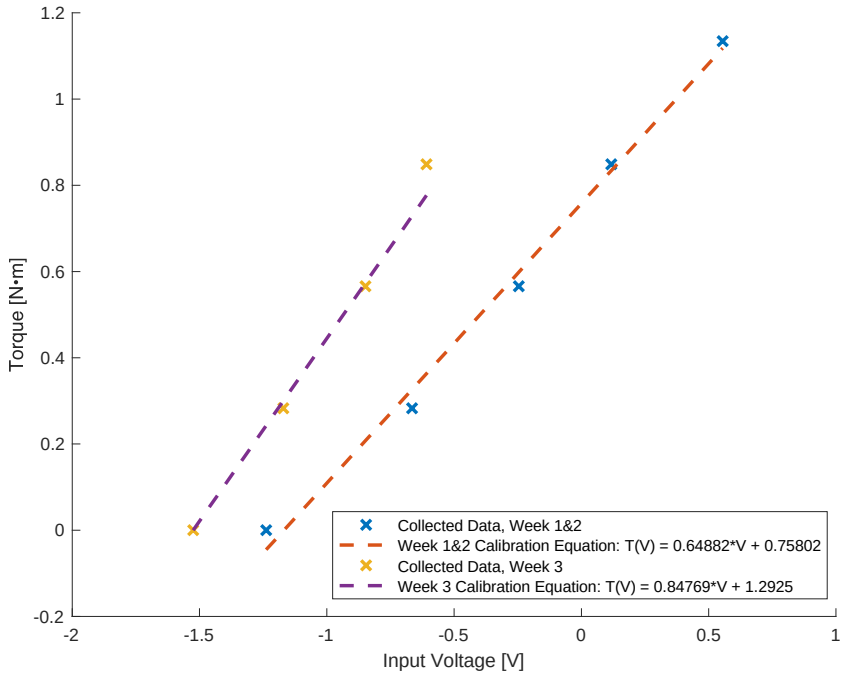


Figure 2: Calibration Data for Torque vs. Input Voltage

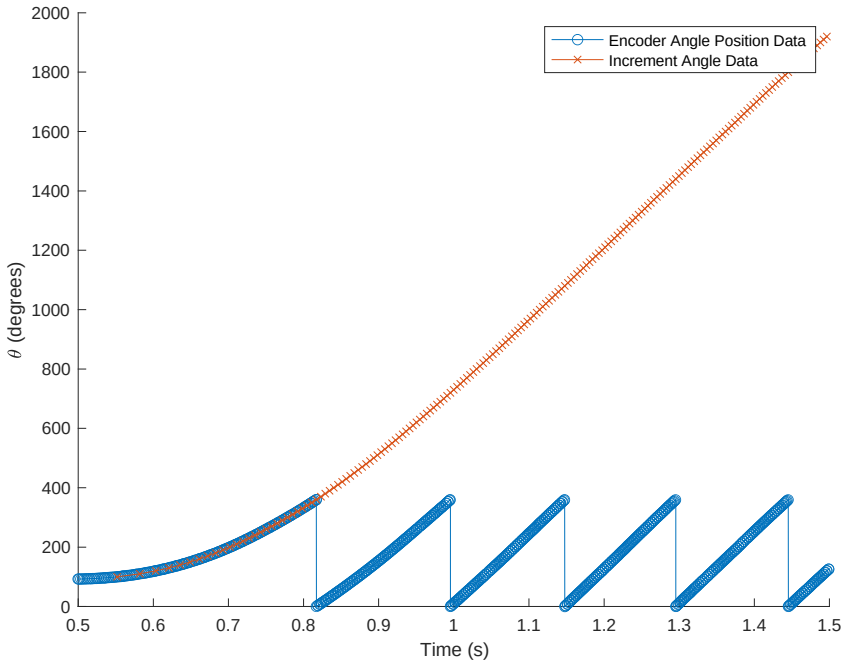


Figure 3: Encoder Angle Position vs. Time for Motor at Low Power (4 Flywheels, No Head)

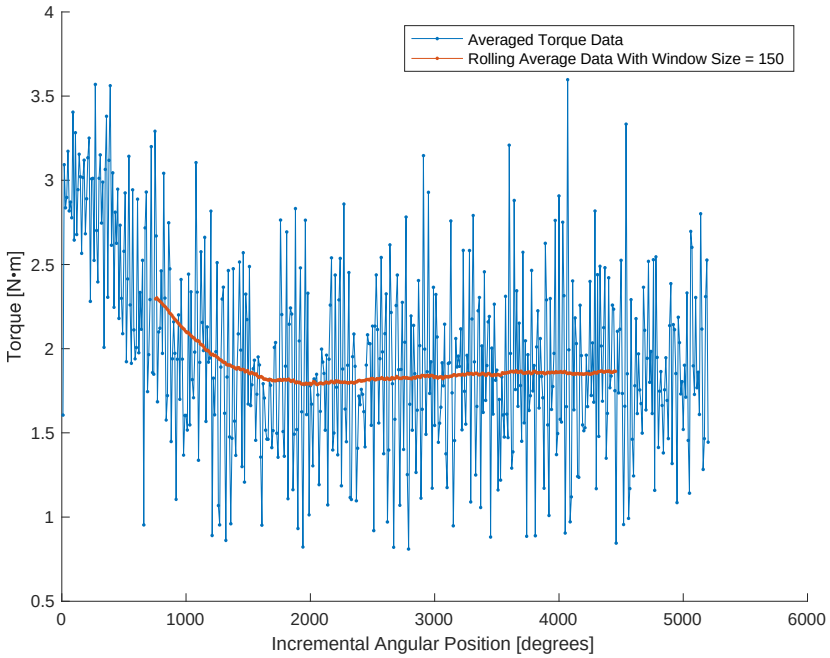


Figure 4: Averaged Torque Data vs. Angular Position for Motor at Low Power (4 Flywheels, No Head)

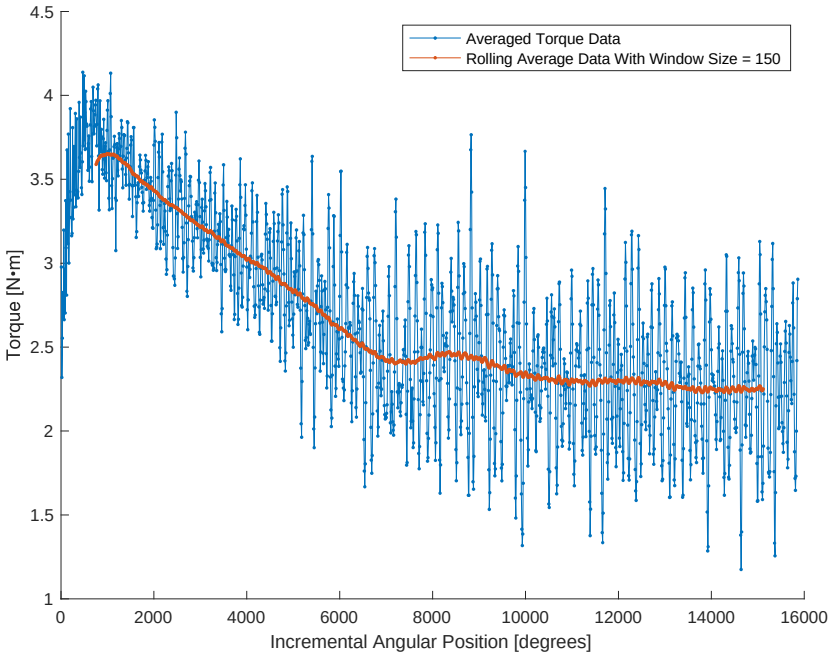


Figure 5: Averaged Torque Data vs. Angular Position for Motor at Medium Power (4 Flywheels, No Heads)

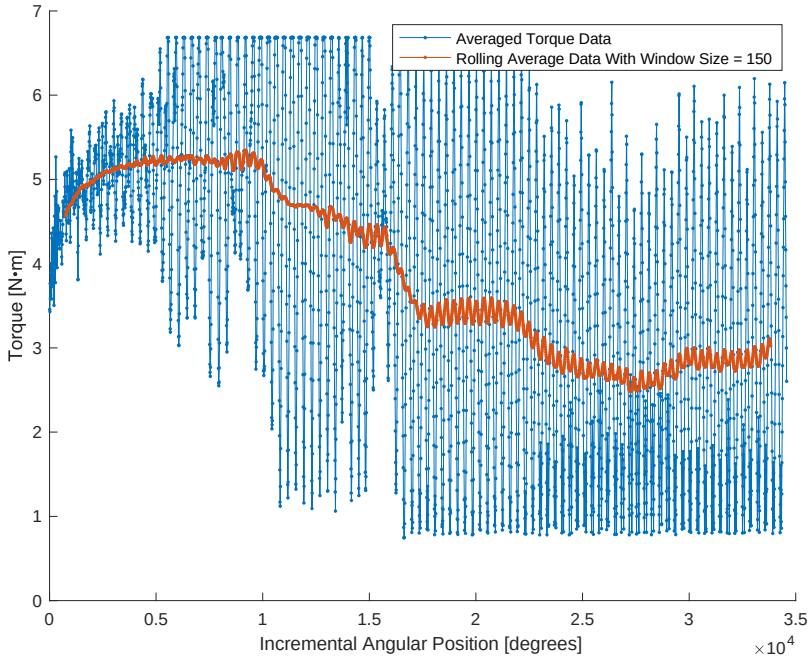


Figure 6: Averaged Torque Data vs. Angular Position for Motor at High Power (4 Flywheels, No Head)

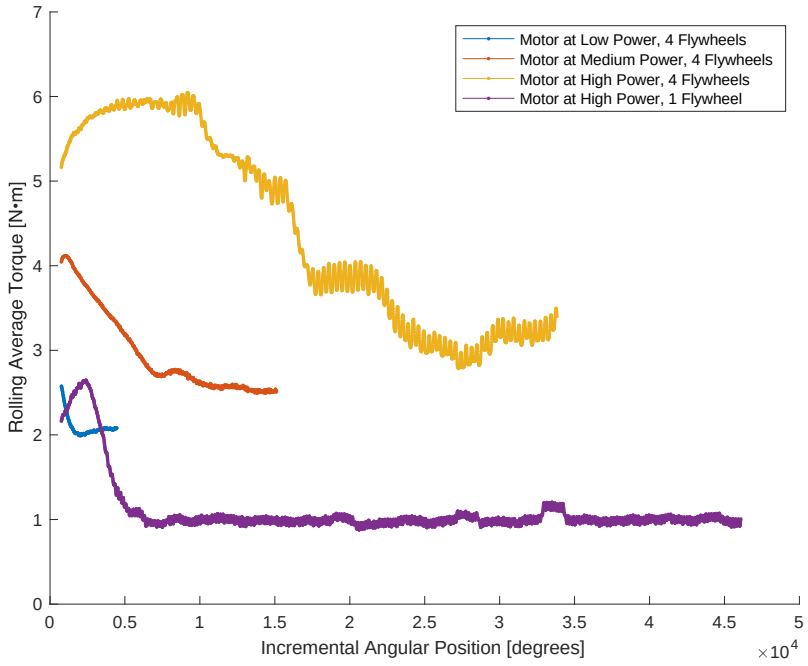


Figure 7: Rolling Average Torque vs. Incremental Angular Position (Low, Medium, and High Power with 4 Flywheels, and High Power with 1 Flywheel, No Head)

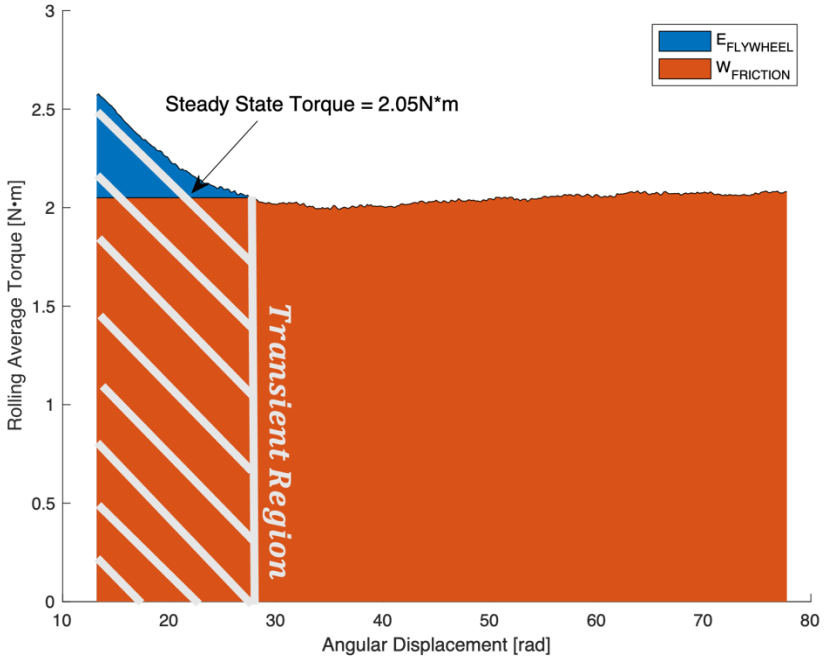


Figure 8: Work Done by Motor and Friction at Low Power (4 Flywheels, No Head)

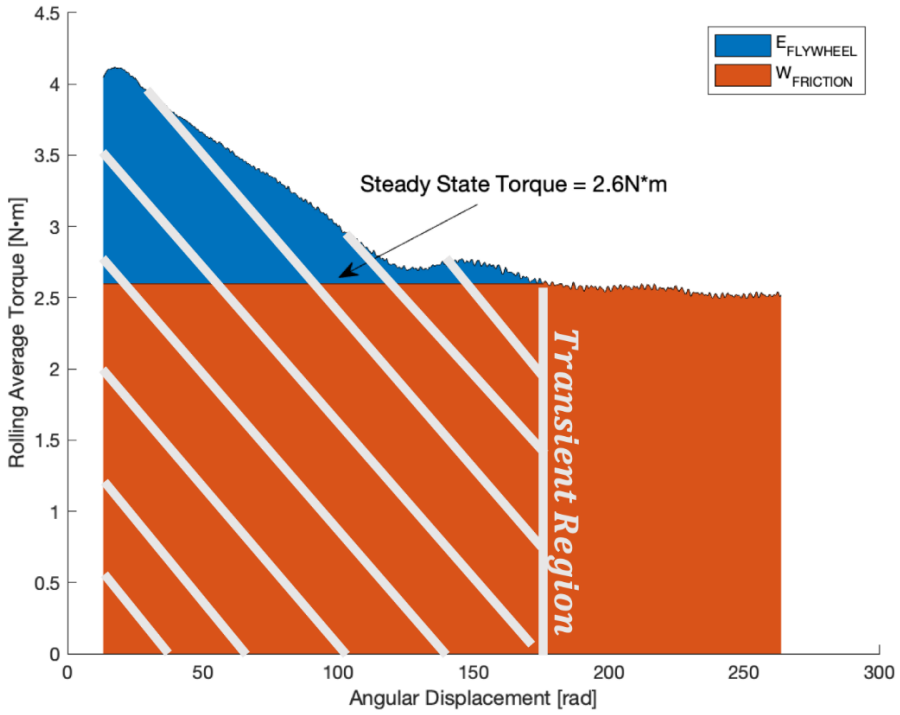


Figure 9: Work Done by Motor and Friction at Medium Power (4 Flywheels, No Head)

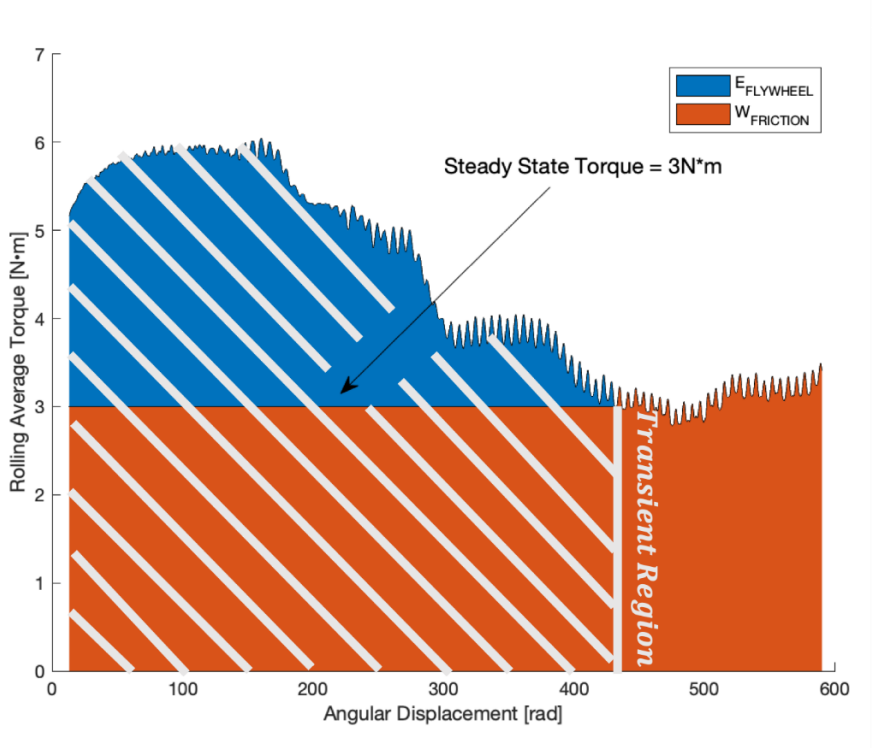


Figure 10: Work Done by Motor and Friction at High Power (4 Flywheels, No Head)

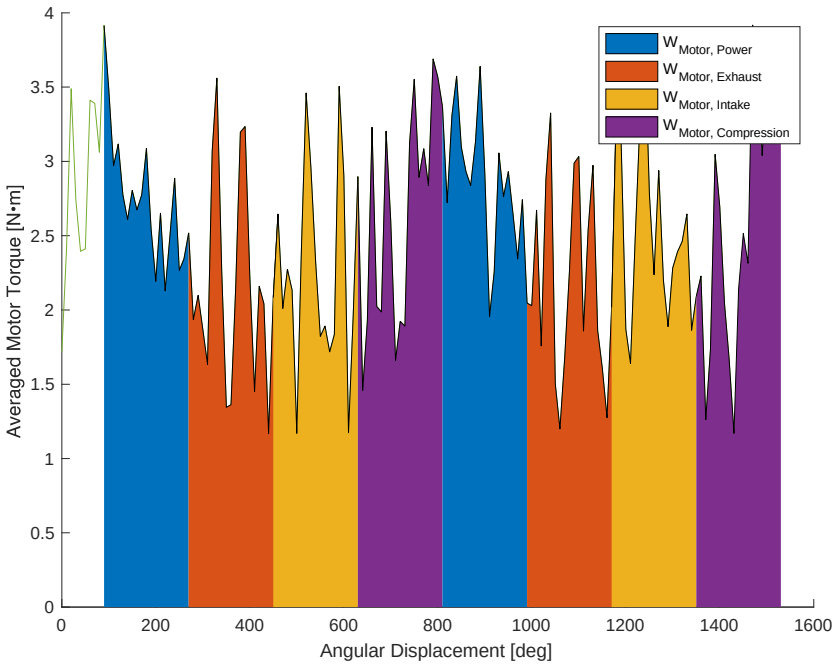


Figure 11: Torque vs. Crank Angle at Low RPM, 4 Flywheels

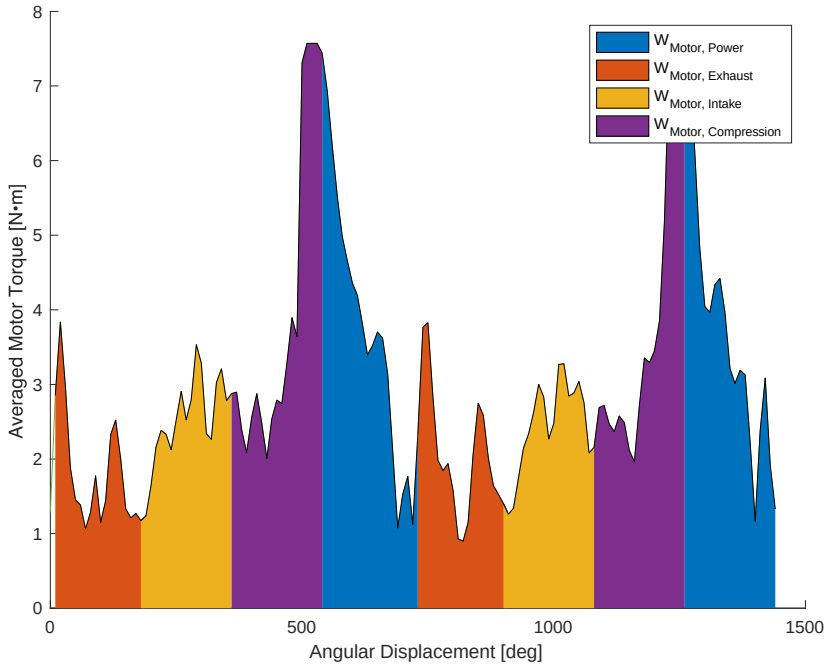


Figure 12: Torque vs. Crank Angle at Low RPM, No Flywheels

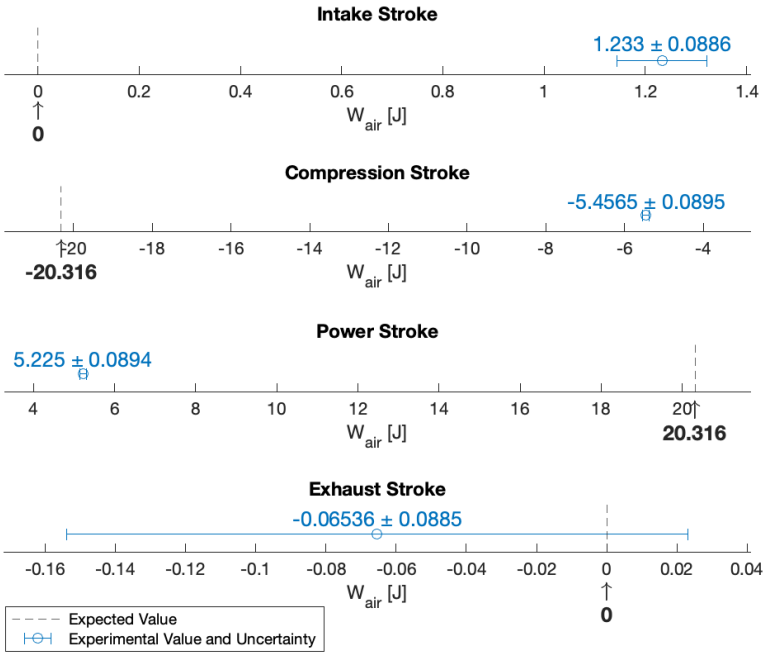


Figure 13: Comparison of W_{air} Experimental and W_{air} Model at each of the 4 Engine Strokes at Low RPM, No Flywheels, No Air Injection

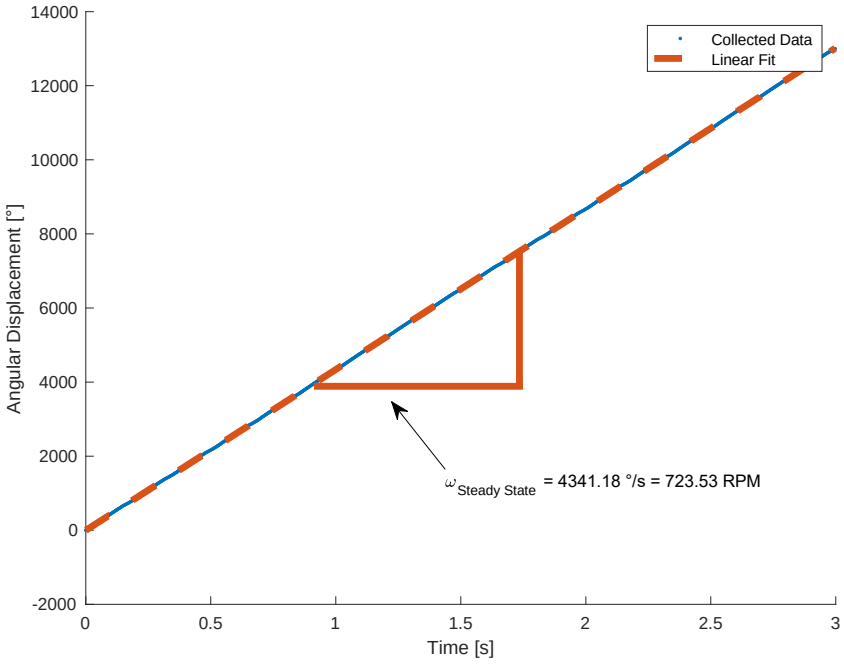


Figure 14: Finding Steady State Angular Velocity Using a Linear Fit with a Configuration of No Flywheels, 95° Injection Start Angle, and 145° Injection End Angle

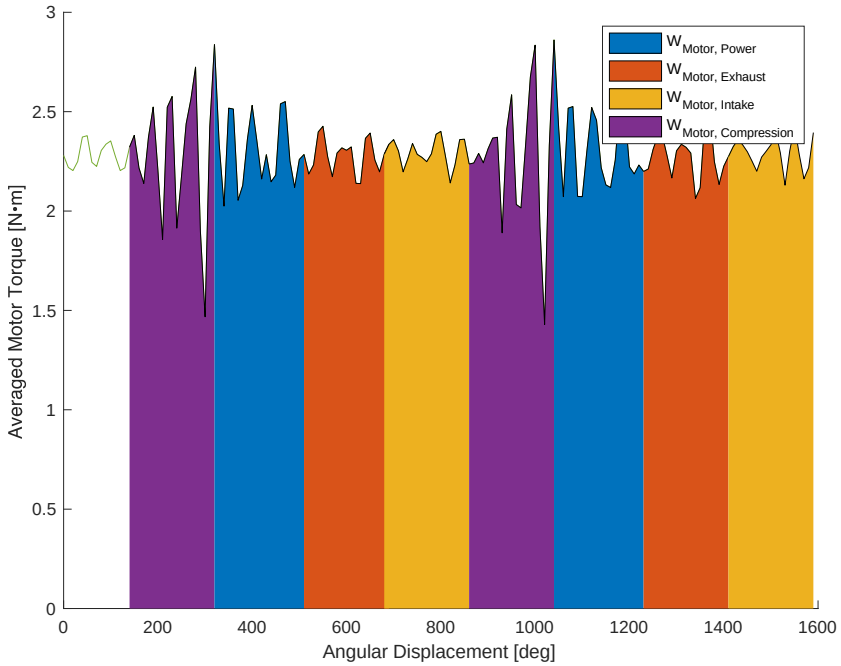


Figure 15: Averaged Torque vs. Angular Displacement at Steady State Angular Velocity with No Flywheels, 100° Injection Start Angle, and 155° Injection End Angle

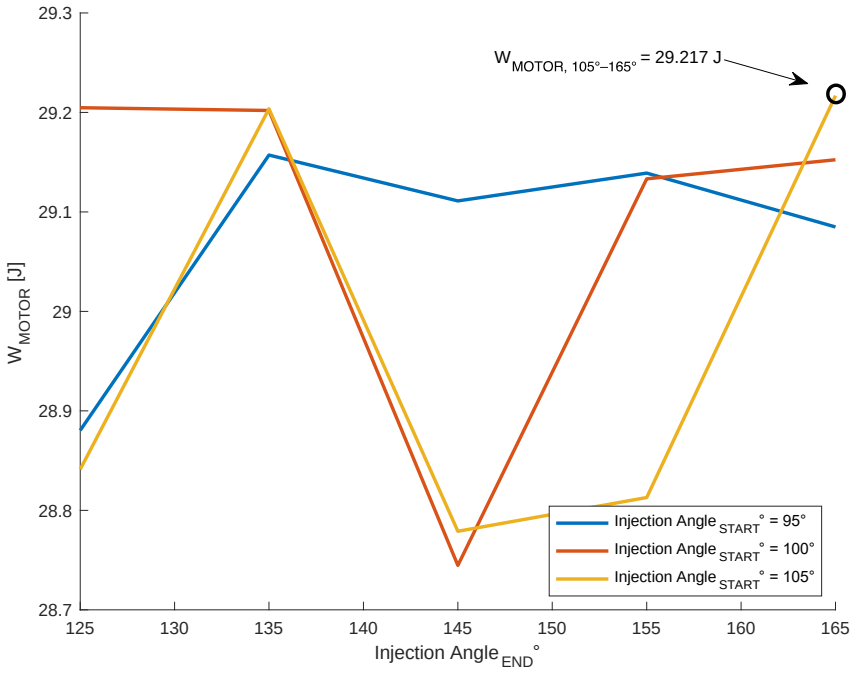


Figure 16: Summation of W_{motor} Along All 4 Strokes for Various Start and End Air Injection Angles

Tables:

Table 1: Experiment 1 Results With Head Off, No Air Injection

	Low Speed Setting, 4 Flywheels	Medium Speed Setting, 4 Flywheels	High Speed Setting, 4 Flywheels	High Speed Setting, 1 Flywheel
$\omega_{STEADY\ STATE}$ [rad/s]	39.0867	138.474	372.823	240.91
$\omega_{STEADY\ STATE}$ [rpm]	373.25	1322.3	3560.2	2300.5
W_{MOTOR} [J]	199.46 ± 0.303	828.99 ± 0.490	2656.5 ± 0.739	909.735 ± 0.769
$W_{FRICTION}$ [J]	132.08	647.432	1748	786.668
$I_{FLY,\ EXP}$ [kg•m^2]	0.0882	0.0189	0.0131	0.00424
$I_{FLY,\ THEORY}$ [kg•m^2]	0.0098	0.0098	0.0098	0.0025
P_{MOTOR} [W]	66.487	276.44	885.5	181.947

Table 2: W_{motor} for Each Stroke at Various Engine Configurations with Head On

Configuration	$\Omega_{STEADY\ STATE}$ [RPM]	$W_{MOTOR,\ EXP,\ POWER}$ [J]	$W_{MOTOR,\ EXP,\ EXHAUST}$ [J]	$W_{MOTOR,\ EXP,\ INTAKE}$ [J]	$W_{MOTOR,\ EXP,\ COMPRESSION}$ [J]
Low Power, No Flywheels	549.81	11.6651 ± 0.0894	6.3749 ± 0.0885	7.6732 ± 0.0886	11.8968 ± 0.0895
Medium Power, No Flywheels	1412.04	9.3476 ± 0.0595	8.2492 ± 0.0592	8.6058 ± 0.609	12.0605 ± 0.0602
High Power, No Flywheels	3494.33	11.1902 ± 0.0597	9.6751 ± 0.0592	13.655 ± 0.0604	7.9137 ± 0.0581
Low Power, 4 Flywheels	512.43	8.9968 ± 0.0856	7.3613 ± 0.0847	7.9198 ± 0.0854	8.1628 ± 0.0854
Medium Power, 4 Flywheels	1371.92	9.1102 ± 0.0579	7.4919 ± 0.0566	8.510 ± 0.0568	8.034 ± 0.0567

Table 3: W_{air} at Various Start and End Air Injection Angles with Head On, No Flywheels

$\theta_{START}° - \theta_{END}°$	$\omega_{STEADY\ STATE}$ [RPM]	$W_{AIR,\ EXP,\ POWER}$ [J]	$W_{AIR,\ EXP,\ EXHAUST}$ [J]	$W_{AIR,\ EXP,\ INTAKE}$ [J]	$W_{AIR,\ EXP,\ COMPRESSION}$ [J]
95° – 125°	668.37	0.275 ± 1.75	-0.201 ± 1.75	0.242 ± 1.75	-0.251 ± 1.75
95° – 135°	718.85	0.141 ± 1.75	-0.089 ± 1.75	0.112 ± 1.80	-0.144 ± 1.75

95° – 145°	723.53	0.107 ± 1.80	-0.115 ± 1.75	0.079 ± 1.75	-0.105 ± 1.75
95° – 155°	721.21	0.131 ± 1.80	-0.096 ± 1.75	0.099 ± 1.75	-0.126 ± 1.75
95° – 165°	713.62	0.136 ± 1.75	-0.094 ± 1.80	0.120 ± 1.75	-0.102 ± 1.75
100° – 125°	567.40	0.454 ± 1.75	-0.412 ± 1.80	0.329 ± 1.75	-0.458 ± 1.75
100° – 135°	683.21	0.246 ± 1.75	-0.186 ± 1.80	0.172 ± 1.75	-0.191 ± 1.75
100° – 145°	704.51	0.259 ± 1.75	-0.120 ± 1.75	0.139 ± 1.75	-0.053 ± 1.75
100° – 155°	700.40	0.237 ± 1.80	-0.111 ± 1.75	0.159 ± 1.75	-0.091 ± 1.75
100° – 165°	692.51	0.120 ± 1.75	-0.132 ± 1.80	0.211 ± 1.75	-0.213 ± 1.75
105° – 125°	534.57	0.497 ± 1.75	-0.478 ± 1.75	0.452 ± 1.75	-0.478 ± 1.75
105° – 135°	649.67	0.308 ± 1.75	-0.207 ± 1.80	0.264 ± 1.75	-0.266 ± 1.75
105° – 145°	665.97	0.276 ± 1.75	-0.161 ± 1.75	0.252 ± 1.75	-0.197 ± 1.75
105 – 155°	669.94	0.281 ± 1.75	-0.167 ± 1.75	0.248 ± 1.75	-0.195 ± 1.75
105° – 165°	666.87	0.282 ± 1.75	-0.219 ± 1.80	0.226 ± 1.75	-0.203 ± 1.75

Table 4: $W_{air,\ model}$ and $W_{air\ discrepancy}$ at Various Start and End Air Injection Angles with Head On, No Flywheels

$\theta_{START}° - \theta_{END}°$	$W_{AIR,\ MOD,\ POWER}$ [J]	$W_{AIR,\ MOD,\ EXHAUST}$ [J]	$W_{AIR,\ MOD,\ INTAKE}$ [J]	$W_{AIR,\ MOD,\ COMPRESSION}$ [J]	$W_{DISC,\ POWER}$ [J]	$W_{DISC,\ EXHAUST}$ [J]	$W_{DISC,\ INTAKE}$ [J]	$W_{DISC,\ COMPRESSION}$ [J]
95° – 125°	35.384	0	0	-20.316	-35.109	-0.201	0.242	20.065
95° – 135°	36.106	0	0	-20.316	-35.965	-0.089	0.112	20.172
95° – 145°	36.498	0	0	-20.316	-36.391	-0.115	0.079	20.211
95° – 155°	36.672	0	0	-20.316	-36.541	-0.096	0.099	20.19
95° – 165°	36.727	0	0	-20.316	-36.591	-0.094	0.12	20.214
100° – 125°	34.010	0	0	-20.316	-33.556	-0.412	0.329	19.858
100° – 135°	34.731	0	0	-20.316	-34.485	-0.186	0.172	20.125
100° – 145°	35.123	0	0	-20.316	-34.864	-0.12	0.139	20.263
100° – 155°	35.297	0	0	-20.316	-35.06	-0.111	0.159	20.225
100° – 165°	35.352	0	0	-20.316	-35.232	-0.132	0.211	20.103
105° – 125°	32.605	0	0	-20.316	-32.108	-0.478	0.452	19.838
105° – 135°	33.327	0	0	-20.316	-33.019	-0.207	0.264	20.05
105° – 145°	33.719	0	0	-20.316	-33.443	-0.161	0.252	20.119
105 – 155°	33.893	0	0	-20.316	-33.612	-0.167	0.248	20.121
105° – 165°	33.948	0	0	-20.316	-33.666	-0.219	0.226	20.113

Equations:

Equation 1: Work-Energy Balance Equation of the Internal Combustion Engine

$$\boldsymbol{E_{fly,b} - E_{fly,a} = W_{motor,a\to b} + W_{friction,a\to b} + W_{piston,a\to b} + W_{air,a\to b} + W_{error,a\to b}}$$

Equation 2: Work-Energy Balance Equation of the Internal Combustion Engine with Head Off, Powered by Motor

$$\boldsymbol{E_{fly,b} - E_{fly,a} = W_{motor,a\to b} + W_{friction,a\to b}}$$

$$W_{motor} = \int T_{motor}(\theta)d\theta$$

$$W_{frict} = \int T_{ss}\, d\theta$$

$$E_{fly} = \frac{1}{2} I_{fly}\omega^2$$

$$I_{fly} = \frac{1}{2} MR^2$$

$$P = \frac{W}{t}$$

Equation 3: Experimental Work Due to Air. W_{piston} is considered negligeable.

$$\boldsymbol{W_{air,a\to b} = -W_{motor,a\to b} - W_{friction,a\to b} - \cancel{W_{piston}} + \frac{1}{2} I_{fly}(\omega_{fly,b}^2 - \omega_{fly,a}^2)}$$

Equation 4: W_{air} for the Model at Different Engine Strokes

$$W_{air,model,intake} = W_{air,model,exhaust} = 0$$

$$W_{air,model,compression} = \int_a^b Pdv = \frac{P_a(V_a)^\gamma}{1-\gamma}\left(V_b^{1-\gamma} - V_a^{1-\gamma}\right) - P_{atm}(V_b - V_a)$$

$$\begin{aligned} W_{air,model,power} &= W_{air,model,power,1\to 2,adiabatic} + W_{air,model,power,2\to 3,isobaric} \\ &+ W_{air,model,power,3\to 4,adiabatic} \end{aligned}$$

$$W_{air,model,power,1\to 2,adiabatic} = \int_1^2 Pdv = \int_1^2 \frac{P_1(V_1)^\gamma}{1-\gamma}\left(V_2^{1-\gamma} - V_1^{1-\gamma}\right) - P_{atm}(V_2 - V_1)$$

$$W_{air,model,power,2\to 3,isobaric} = \int_2^3 Pdv = \int_2^3 \left(P_{line,abs} - P_{atm}\right)$$

$$W_{air,model,power,3\to4,adiabatic} = \int_3^4 Pdv = \int_3^4 \frac{P_3(V_3)^\gamma}{1-\gamma}\left(V_4^{1-\gamma} - V_3^{1-\gamma}\right) - P_{atm}(V_4 - V_3)$$

Equation 5: Finding Uncertainty of Motor Work

$$\boldsymbol{U_{c,Wmotor} = \sqrt{m} \cdot U_{c,Wi}}$$

$$m = number\ of\ bins$$

$$U_{c,Wi} = \sqrt{\left(\frac{\partial W_i}{\partial \Delta\theta_i} \cdot U_{c,\Delta\theta_i}\right)^2 + \left(\frac{\partial W_i}{\partial T_{avg,i}} \cdot U_{c,T_{avg,i}}\right)^2}$$

$$U_{c,\Delta\theta_i} = \frac{\pi}{180°} \cdot \frac{360°}{4000} \cdot \sqrt{\left(U_{a,e,count}\right)^2 + \left(U_{p,e,count}\right)^2}$$

$$U_{c,Tavg,i} = \sqrt{\left(U_{a,Tavg,i}\right)^2 + \left(U_{p,Tavg,i}\right)^2}$$

Equation 6: Calculating Discrepancy Work Betweem the Experiment and Model

$$W_{discrepancy} = \left(W_{air,a\to b}\right)_{experiment} - \left(W_{air,a\to b}\right)_{model}$$

References:

[1] Z. Cloutier, *Lecture 09 – ME 4056 Systems Lab (IC Engines 1)*, Atlanta: Georgia Institute of Technology, 2022.

[2] *IC Engine Guideline*, Spring 2022. Georgia Institute of Technology, Atl., Ga., U.S.A., Mar 2022.